

EEE4114F: Digital Signal Processing

Class Test

19 March 2018

SOLUTIONS

Name:

Student number:

Information

- The test is closed-book.
 - This test has *four* questions, totalling 20 marks.
 - Answer *all* the questions.
 - You have 45 minutes.
 - An information sheet is attached.
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1. (5 marks) The input signal

$$x[n] = \begin{cases} 1 & 0 \leq n \leq 9 \\ 0 & \text{otherwise} \end{cases}$$

is applied to a system with impulse response

$$h[n] = (1/3)^n u[n].$$

Find and plot the values of the output signal $y[n]$ over the range $n = -4$ to $n = 4$.

Since the input can be written as $x[n] = u[n] - u[n - 10]$, the output will be

$$y[n] = h[n] * x[n] = h[n] * (u[n] - u[n - 10]) = g[n] - g[n - 10]$$

with $g[n] = h[n] * u[n]$.

Now

$$g[n] = \sum_{k=-\infty}^{\infty} h[k] u[n - k] = \sum_{k=-\infty}^n h[k],$$

where the sum is truncated because $u[n - k] = 0$ for $n - k < 0$, or $k > n$. Since $h[k] = 0$ for $k < 0$ the output $g[n] = 0$ for $n < 0$. For $n > 0$ we have

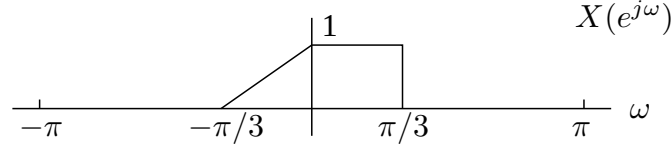
$$g[n] = \sum_{k=0}^n (1/3)^k = \frac{1 - (1/3)^{n+1}}{1 - 1/3}.$$

The full step response is therefore

$$g[n] = \frac{3}{2} (1 - (1/3)^{n+1}) u[n].$$

Since $g[n - 10] = 0$ for $n < 10$ it has no effect on the output values over the range specified, so it is easy to find and plot the required values of $y[n]$.

2. (5 marks) A sequence $x[n]$ has a zero-phase DTFT $X(e^{j\omega})$ given below:



Sketch the magnitude and phase of the DTFT of the following sequences:

(a) $y_1[n] = x[n - 1]$.

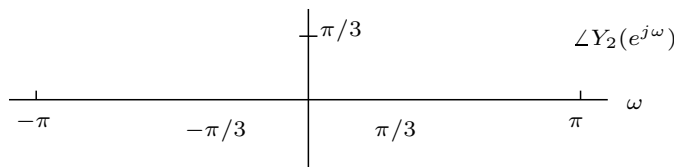
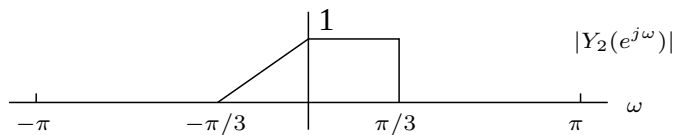
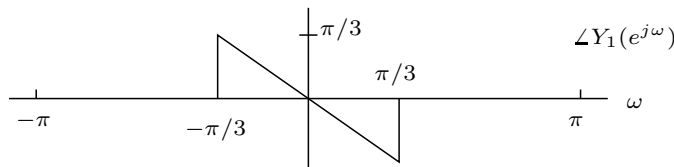
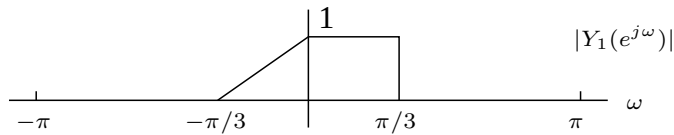
(b) $y_2[n] = x^*[-n]$.

- (a) The transform satisfies $Y_1(e^{j\omega}) = e^{-j\omega} X(e^{j\omega})$. Since $X(e^{j\omega})$ is zero phase we have $|Y_1(e^{j\omega})| = X(e^{j\omega})$ and $\angle Y_1(e^{j\omega}) = -\omega$, plotted below.

- (b) The transform is as follows:

$$\begin{aligned} Y_2(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x^*[-n] e^{-j\omega n} = \sum_{m=-\infty}^{\infty} x^*[m] e^{j\omega m} \\ &= \left(\sum_{m=-\infty}^{\infty} x[m] e^{-j\omega m} \right)^* = X^*(e^{j\omega}) = X(e^{j\omega}), \end{aligned}$$

where the last step is true because $X(e^{j\omega})$ is real.



3. (5 marks) Consider the discrete LTI system represented by

$$y[n] = x[n] - x[n-1]$$

where $x[n]$ and $y[n]$ are the input and output respectively.

- (a) Determine and plot the impulse response $h[n]$. Is the system stable?
 - (b) Determine and plot the step response corresponding to $x[n] = u[n]$.
 - (c) Find $H(e^{j\omega})$ and plot its magnitude.
 - (d) Determine and plot the response to the input $x[n] = (-1)^n$.
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- (a) The impulse response $h[n]$ is the output when the input is $x[n] = \delta[n]$, so $h[n] = \delta[n] - \delta[n-1]$. Since $\sum_{n=-\infty}^{\infty} |h[n]| = 2 < \infty$ the system is stable.
 - (b) When the input is $x[n] = u[n]$ then the output will be $y[n] = u[n] - u[n-1] = \delta[n]$.
 - (c) The frequency response is

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} (\delta[n] - \delta[n-1]) e^{-j\omega n} = 1 - e^{-j\omega} = e^{-j\omega/2} (e^{j\omega/2} - e^{-j\omega/2}) \\ &= 2je^{-j\omega/2} \sin(\omega/2), \end{aligned}$$

so $|H(e^{j\omega})| = 2|\sin(\omega/2)|$.

- (d) The response to $x[n] = (-1)^n = e^{j\pi n}$ will be $y[n] = H(e^{j\pi})e^{j\pi n} = 2e^{j\pi n} = 2(-1)^n$.

4. (5 marks) A system has impulse response $h[n] = (1/2)^n u[n]$. Determine the input $x[n]$ to the system if the output is given by $y[n] = 2\delta[n - 4]$. Explicitly state regions of convergence for any Z-transforms.

The system function is

$$H(z) = \frac{1}{1 - 1/2z^{-1}}$$

with ROC $|z| > 1/2$. The output is

$$Y(z) = 2z^{-4}$$

for all z . The input can then be found to be

$$X(z) = \frac{Y(z)}{H(z)} = 2z^{-4}(1 - 1/2z^{-1}) = 2z^{-4} - z^{-5}$$

with ROC the entire z -plane, so the input is

$$x[n] = 2\delta[n - 4] - \delta[n - 5].$$

Discrete-time Fourier transform properties

Sequences $x[n]$, $y[n]$	Transforms $X(e^{j\omega})$, $Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{-j\omega}) Y(e^{-j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) Y(e^{j(\omega - \theta)}) d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi \delta(\omega + 2\pi k)$
$a^n u[n]$ $(a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi \delta(\omega + 2\pi k)$
$(n + 1)a^n u[n]$ $(a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n]$, $y[n]$	Transforms $X(z)$, $Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$-u[-n - 1]$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
$\delta[n - m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1 - az^{-1}}$	$ z > a $
$-a^n u[-n - 1]$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
$-na^n u[-n - 1]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N - 1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1 - a^N z^{-N}}{1 - az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1 - \cos(\omega_0) z^{-1}}{1 - 2 \cos(\omega_0) z^{-1} + z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1 - r \cos(\omega_0) z^{-1}}{1 - 2r \cos(\omega_0) z^{-1} + r^2 z^{-2}}$	$ z > r $

